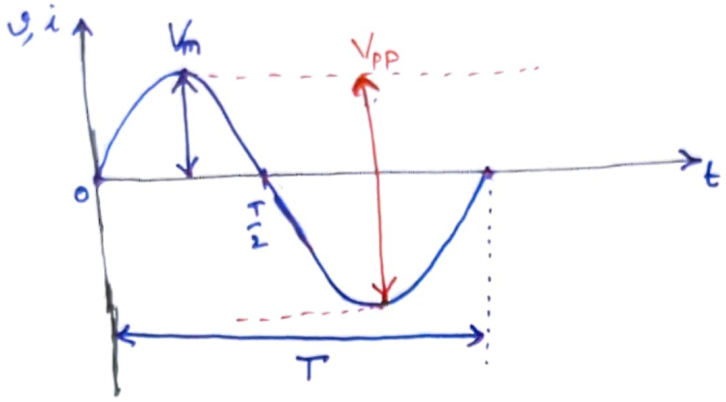
UNIT-2

Sinusoids :



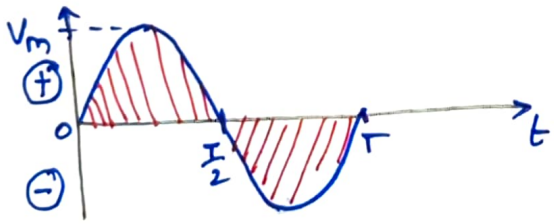
V_m = Peak value

V_{p-p} = Peak - Peak value

T = Period of the sine wave

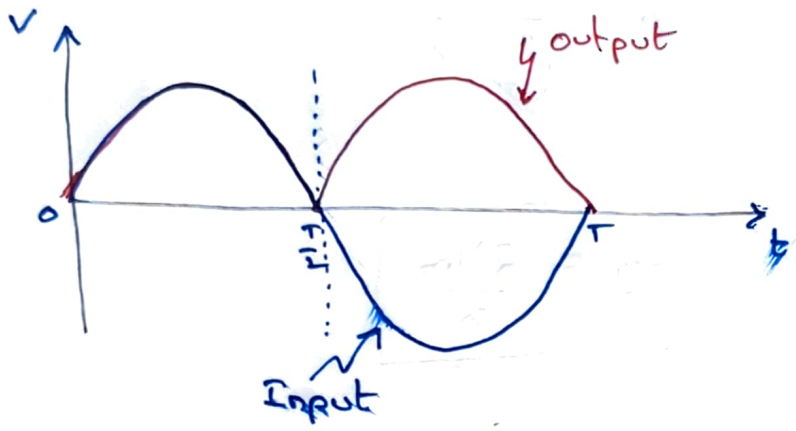
Frequency = $\frac{1}{T}$ Hz (Hertz)

Q. What is average or mean value of sinusoid?



$$V_{avg} = \frac{\text{area under the curve}}{\text{time base } (T)} = \frac{0}{T} = 0$$

Example: What is the average or mean value of output voltage of full wave diode bridge rectifier?



Sol:- The average value of output of full wave rectifier is given as below

$$V_{avg} = \frac{\text{area under "RED" curve}}{(T/2)}$$

$$\text{Input} = V_m \sin(\omega t)$$

ω → $2\pi f$ or $\frac{2\pi}{T}$

$$\therefore V_{avg} = \frac{1}{(T/2)} \int_0^{T/2} v \, dt$$

$$= \frac{2}{T} \left[\int_0^{T/2} V_m \cdot \sin(\omega t) \, dt \right]$$

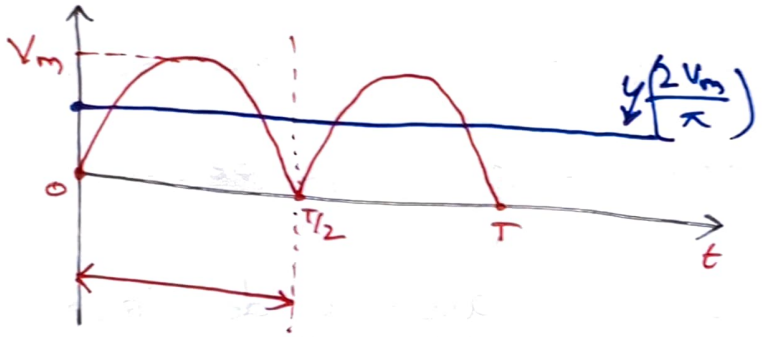
$$= \frac{2V_m}{T} \left[-\frac{\cos(\omega t)}{\omega} \right]_0^{T/2}$$

$$= \frac{2V_m}{\omega T} \left[-\cos\left(\frac{2\pi}{T} \cdot \frac{T}{2}\right) + \cos\left(\frac{2\pi}{T} \cdot 0\right) \right]$$

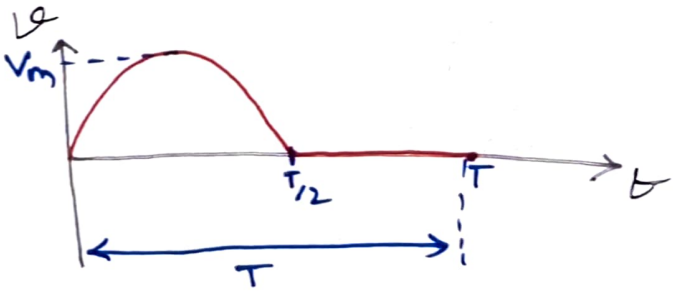
$$\therefore \quad V_{avg} = \frac{2V_m}{\omega T}\left[1+1\right]$$

$$\Rightarrow \quad V_{avg} = \frac{4V_m}{\left(\frac{2\pi}{T}\right)T}$$

$$\Rightarrow \quad \boxed{V_{avg} = \frac{2V_m}{\pi}}$$



Example 02: Obtain the average value of the given sinusoidal waveform.



$$V_{avg} = \frac{1}{T}\int_0^{T/2} V_m \sin(\omega t)\cdot dt$$

$$= \frac{1}{T}\cdot V_m \cdot \left[-\frac{\cos(\omega t)}{\omega}\right]_0^{T/2}$$

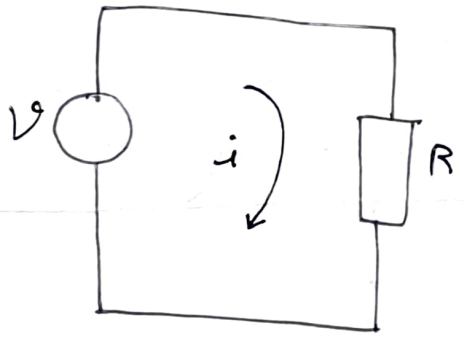
$$\therefore \quad \boxed{V_{avg} = \frac{V_m}{\pi}}$$

Root Mean Square (RMS) value of sinusoid :

→ RMS has relation to power.

→ RMS is that quantity of voltage or current which produces the power.

→ Firstly analyze the power in the following circuit.

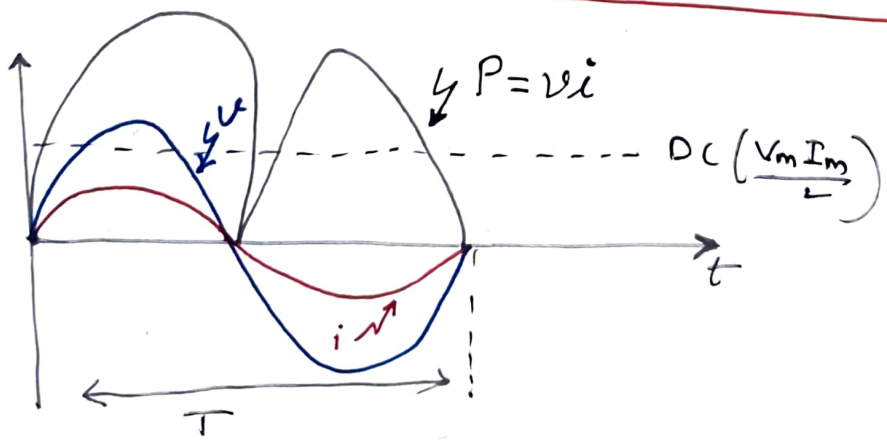


$$v = V_m \sin(\omega t)$$

$$i = \frac{v}{R} = \frac{V_m \sin(\omega t)}{R}$$

$$I_m = \frac{V_m}{R}$$

$$\boxed{i = I_m \sin(\omega t)}$$



$\therefore$ power = $v \times i$ (watts)

$\Rightarrow$ Volts x Amps = Watts

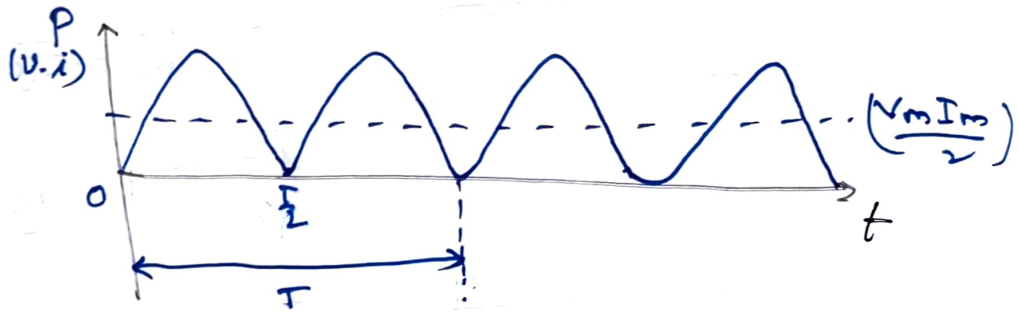
$\therefore P = vi$

$= V_m \sin(\omega t) \cdot I_m \sin(\omega t)$

$= V_m I_m \sin^2(\omega t)$ $\quad \left[\sin^2(\theta) = \frac{1-\cos 2\theta}{2} \right]$

$= V_m I_m \left[\frac{1-\cos 2\theta}{2} \right]$

$P = \underbrace{\frac{V_m I_m}{2}}_{\text{Dc term}} - \underbrace{\frac{V_m I_m}{2} \cos(2\omega t)}_{\text{AC term with twice frequency}}$



$\therefore P_{(avg)} = \frac{1}{T} \int_0^T v.i \, dt$

$= \frac{1}{T} \int_0^T \frac{V_m I_m}{2} \left(1 - \cos(2\omega t)\right) dt$

$= \frac{V_m I_m}{2T} \int_0^T (1 - \cos 2\omega t) \, dt$

$$= \frac{V_m I_m}{2\pi}\left[[t]_0^T - \left[\frac{\sin(2\omega t)}{2\omega}\right]_0^T \right]$$

$$\boxed{P_{avg} = \frac{V_m I_m}{2}}$$

But we know that

$$I_m = \frac{V_m}{R}$$

$$\therefore \boxed{P_{avg} = \frac{V_m^2}{2R} = \frac{I_m^2}{2} \cdot R}$$

$$\Rightarrow \text{Amp}^2 \cdot \text{ohms} = \text{watts}$$

$$\Rightarrow \text{Amp} \cdot \underbrace{\text{Amp} \cdot \text{ohm}}_{\text{Volts}}$$

$$\therefore P_{avg} = \left(\frac{I_m}{\sqrt{2}}\right)\left(\frac{I_m}{\sqrt{2}}\right) \cdot R$$

$$\text{Watts} = \underbrace{\text{Amps}}_{\text{RMS}} \quad \underbrace{\text{Amps} \cdot \text{ohms}}_{\text{Volts}}$$

$$= I_{rms} \cdot I_{rms} \cdot R$$

$$= I_{rms}^2 \cdot R$$

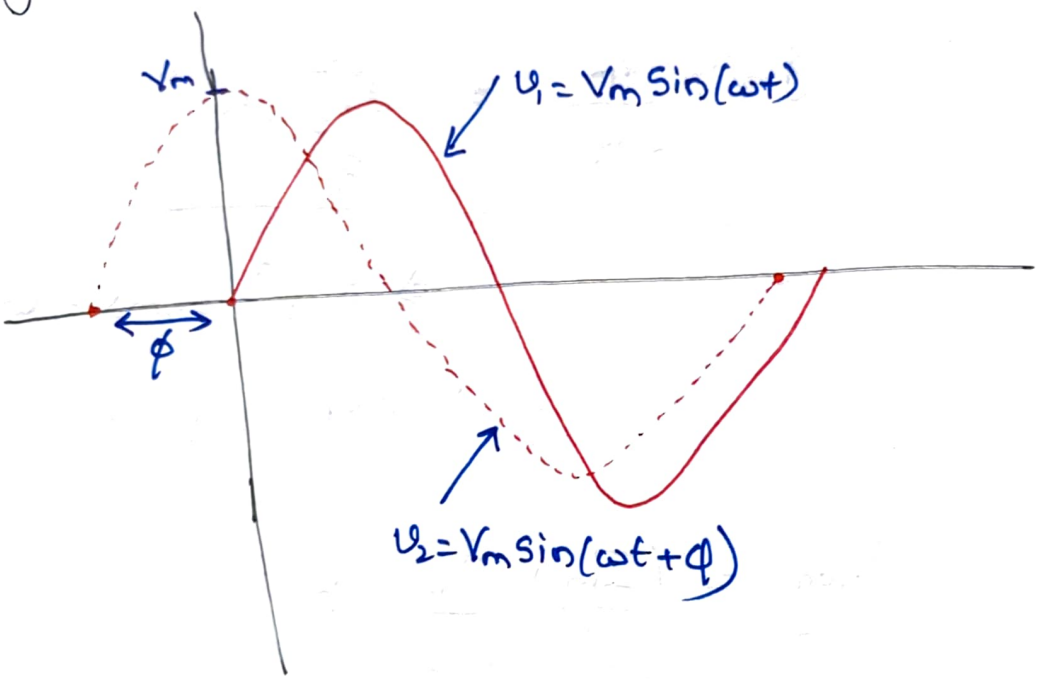
$$\therefore \boxed{I_{rms} = \frac{I_m}{\sqrt{2}}}$$

Consider the following equation:

$v_1(t) = V_m \sin \omega t$ and $v_2(t) = V_m \sin(\omega t + \phi)$

where $(\omega t + \phi)$ is the argument and ϕ is the phase

Both argument and phase can be radians (or) degrees.



→ As observed from the above waveform starting point of v_2 occurs first in time.

→ Thus "v_2 leads v_1" or v_1 lags v_2 by ϕ.

→ If $\phi = 0$ then v_1 and v_2 are "in-phase".

→ If $\phi \neq 0$ then v_1 and v_2 are "out of phase"

Note: In the above waveform v_1 and v_2 are compared because they operate at the same frequency. They do not need to have the same amplitude.

Trignometric Identities:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

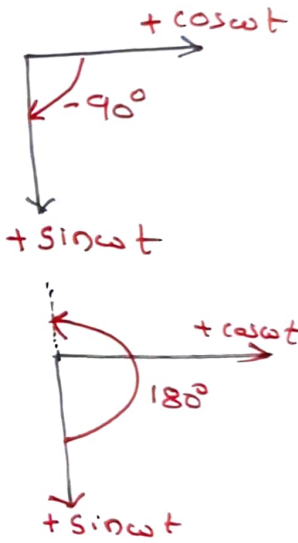
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(\omega t \pm 180^\circ) = -\sin(\omega t)$$

$$\cos(\omega t \pm 180^\circ) = -\cos(\omega t)$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos(\omega t)$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin(\omega t)$$



Example 01:

Find the amplitude, phase, period and frequency of sinusoid:

$$v(t) = 12\cos(50t + 10^\circ)$$

Sol:- $V_m = 12\,V$;

$\phi_{(Phase)} = 10^\circ$;

$\omega = 50\,rad/s$;

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1257\,sec$$

$$f = \frac{1}{T} = 7.958\,Hz$$

Example: 02:

Calculate the phase angle between v_1 and v_2

$v_1 = -10\cos(\omega t + 50^\circ)$ and $v_2 = 12\sin(\omega t - 10^\circ)$.

Stat which sinusoid is leading.

Sol:- In order to compare two sinusoids, both should be of same form.

$$v_1 = -10\cos(\omega t + 150^\circ) = 10\cos(\omega t + 50^\circ - 180^\circ)$$

$$v_1 = 10\cos(\omega t - 130^\circ) \text{ or}$$

$$v_1 = 10\cos(\omega t + 230^\circ) \rightarrow (1)$$

and

$$v_2 = 12\sin(\omega t - 10^\circ) = 12\cos(\omega t - 10^\circ - 90^\circ)$$

$$\Rightarrow v_2 = 12\cos(\omega t - 100^\circ) \rightarrow (2)$$

From (1) & (2) the phase difference is 30°

$$v_2 = 12\cos(\omega t - 130^\circ + 30^\circ)$$

$\therefore v_2$ leads v_1 by 30°

Phasors:

A phasor is a complex number that represents the amplitude and phase of sinusoid.

→ A complex number z can be written in rectangular form as

$$z = x + jy \quad \text{(Rectangular form)}$$

→ The complex number 'z' can also be written in polar or exponential form as

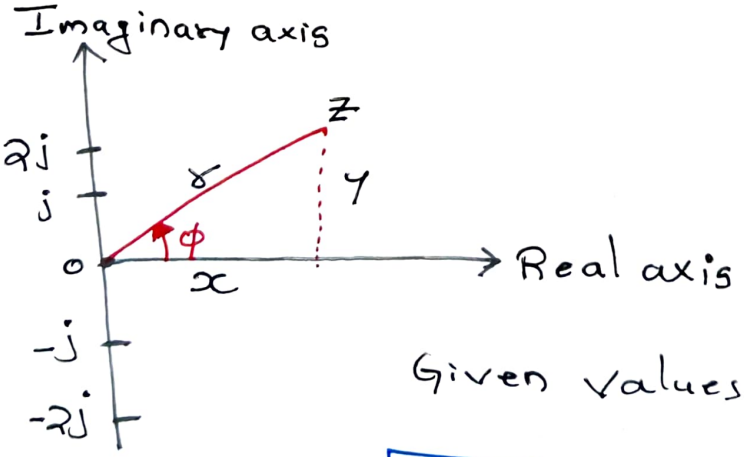
$$z = r\angle\phi = re^{j\phi}$$

(Polar) (exponential)

where

r = magnitude of z

ϕ = phase of z



Given values of x & y

$$r = \sqrt{x^2+y^2}; \quad \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

→ On the other hand if 'r' and 'φ' are known then:

$$x = r\cos\phi \quad \& \quad y = r\sin\phi$$

Thus, z may be written as

$$z = x + jy = r\angle\phi = r[\cos\phi + j\sin\phi]$$

Properties:

Consider two complex numbers

$$z_1 = x_1 + jy_1 = r_1\angle\phi_1 \quad \& \quad z_2 = x_2 + jy_2 = r_2\angle\phi_2$$

(1) $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

(2) $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

(3) $z_1 z_2 = r_1 r_2 \angle\phi_1 + \phi_2$

(4) $\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle\phi_1 - \phi_2$

(5) $\frac{1}{z} = \frac{1}{r}\angle -\phi_1$

(6) $\sqrt{z} = \sqrt{r}\angle\phi/2$

(7) $z^* = x - jy = r\angle -\phi = re^{-j\phi}$

→ The idea of phasor representation is based on Euler's identity.

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

where

$\cos\phi = Re(e^{j\phi})$ → Real part

$\sin\phi = Im(e^{j\phi})$ → Imaginary part

∴ Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$,

$$v(t) = V_m \cos(\omega t + \phi)$$

$$= Re(V_m e^{j(\omega t + \phi)})$$

(or)

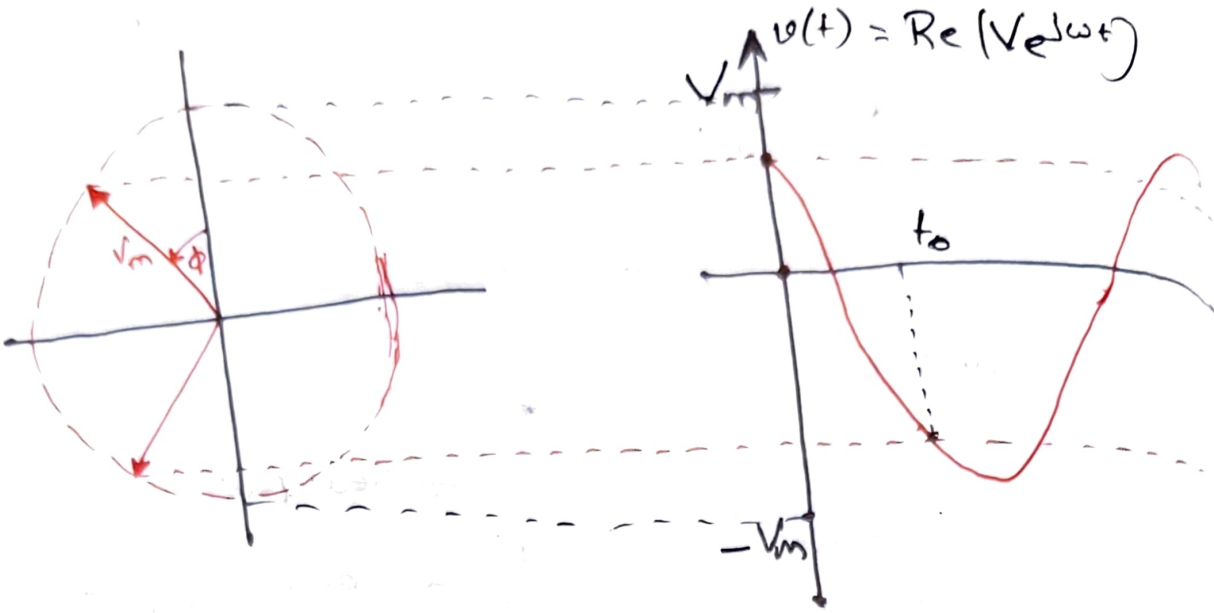
$$v(t) = Re(V_m e^{j\phi} e^{j\omega t})$$

Thus,

$$\boxed{v(t) = Re(V e^{j\omega t})} \rightarrow (a)$$

where

$$\boxed{V = V_m e^{j\phi} = V_m \angle\phi} \rightarrow (b)$$



→ First express sinusoid in cosine form

→ Then it can be written as the real part of a complex number.

→ Take out the factor $e^{j\omega t}$ and whatever left is the phasor corresponding to the sinusoid.

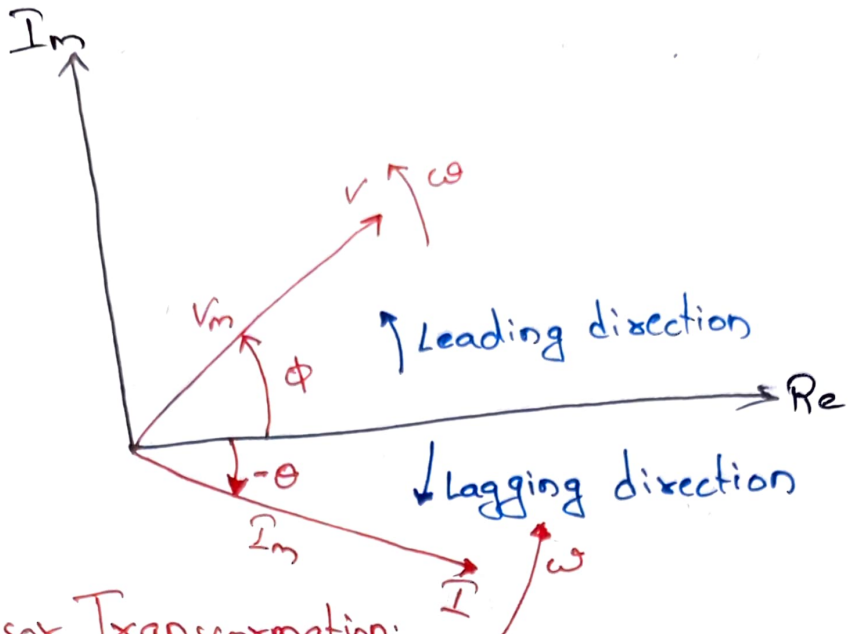
$$v(t) = V_m \cos(\omega t + \phi) \iff V = V_m \angle \phi$$

[Time domain representation] [Phasor representation]

→ Consider two phasors:

$$V = V_m \angle \phi ;$$

$$I = I_m \angle \theta$$



Sinusoid Phasor Transformation:

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90°$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90°$

⇒ The frequency (or time) factor $e^{j\omega t}$ is suppressed, and the frequency is not shown explicity in phasor domain representation because ω is constant.

⇒ However response depends on 'ω'. Thus phasor domain is also known as the "frequency domain".

Now consider the following equation

$$v(t) = Re\left(V e^{j\omega t}\right)$$

$$= V_m \cos(\omega t + \varphi)$$

$$\frac{dv}{dt} = -\omega V_m \sin(\omega t + \varphi)$$

$$= \omega V_m \cos(\omega t + \varphi + 90^\circ)$$

$$= Re\left(\omega V_m e^{j\omega t} e^{j\varphi} e^{j90^\circ}\right)$$

$$= Re\left(j\omega V e^{j\omega t}\right)$$

→ This shows that the derivative $v(t)$ is transformed to phasor domain as $j\omega V$.

$$\frac{dv}{dt} \Longleftrightarrow j\omega V$$

(Time domain) (Phasor domain)

→ Similarly, the integral of $v(t)$ is transformed to phasor domain as $V/j\omega$

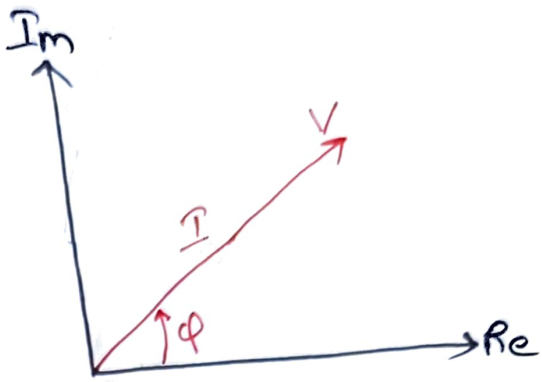
$$\int v \cdot dt \Longleftrightarrow \frac{V}{j\omega}$$

(Time domain) (Phasor domain)

Phasor Relationship for Circuit Elements:

(a)





→ If the current through resistor R is,

$$i = I_m \cos(\omega t + \varphi)$$

→ Then voltage across is given by Ohms law,

$$v = iR = R I_m \cos(\omega t + \varphi)$$

→ The phasor form of this voltage is,

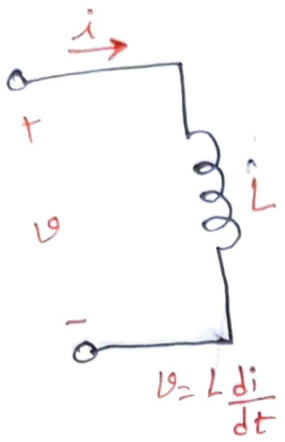
$$V = R \cdot I_m \angle \varphi$$

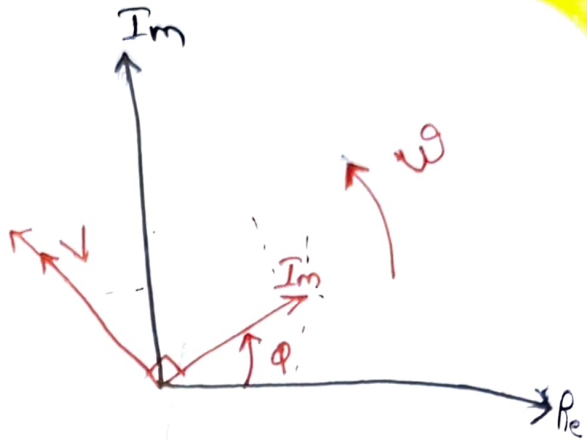
But

$$I = I_m \angle \varphi$$

$$\Rightarrow \boxed{V = I \cdot R}$$

(ii)





For the inductor L, assume the current through it is:

$$i = I_m \cos(\omega t + \varphi)$$

→ The voltage across inductor is

$$v = L\frac{di}{dt} = -\omega L I_m \sin(\omega t + \varphi)$$

⇒ The above equation can be written as

$$v = L I_m \cdot \omega \cos(\omega t + \varphi + 90^\circ)$$

⇒ The phasor transformation can be written as

$$V = \omega L I_m \, e^{j(\varphi + 90^\circ)}$$

$$= \omega L I_m \cdot e^{j\varphi} \cdot e^{j90^\circ}$$

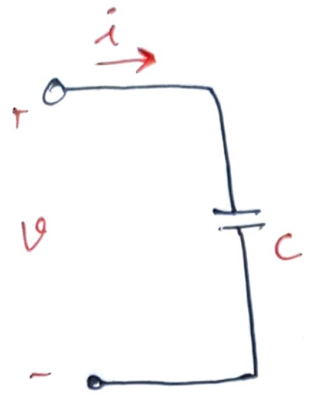
$$= \omega L I_m \angle \phi + 90^\circ$$

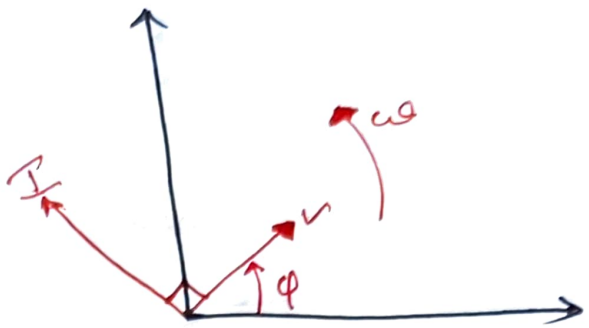
→ But $I_m \angle \varphi = I$ and $e^{j90^\circ} = j$

$$\therefore \boxed{V = j\omega L I}$$

⇒ The current lags the voltage by 90°

(c)





⇒ The current through capacitor is

$$i = C\frac{dv}{dt} \rightarrow (a)$$

$$\rightarrow I = j\omega CV \Rightarrow V = \frac{1}{j\omega C}$$

⇒ Current leads the voltage by 90°.

From (a)

$$\frac{dv_c}{dt} = \frac{d}{dt}(V_m \sin\omega t) = \omega V_m \cos\omega t$$

Therefore

$$i_c = C\cdot\frac{dv_c}{dt} = C(\omega \cdot V_m \cos\omega t)$$

$$= \omega \cdot CV_m \cos\omega t$$

$$\therefore i_c = I_m \sin(\omega t + 90^\circ)$$

where

$$I_m = \omega \cdot CV_m$$

Example: The voltage across a resistor is indicated. Find the sinusoidal expression for the current if the resistor is 10Ω. Sketch the curves for v & i.

(a) $v = 100 \sin 377t$

(b) $v = 25 \sin(377t + 60^\circ)$

Sol:- (a) $I_m = \frac{V_m}{R} = \frac{100\,V}{10\,\Omega} = 10\,A$

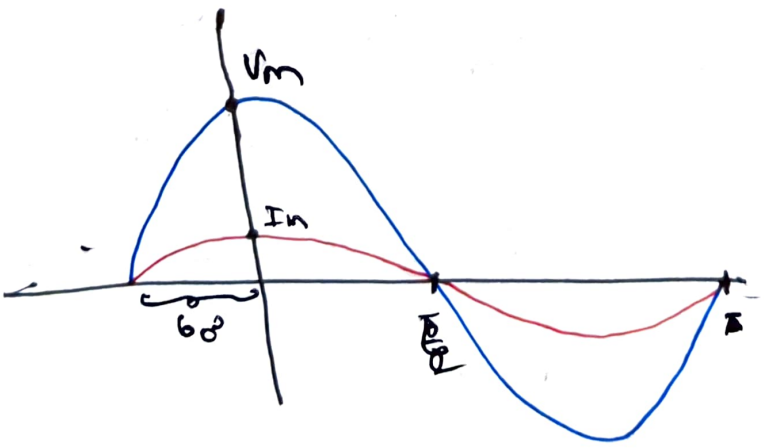
v & i are in phase,

$\therefore i = 10 \sin 377t$

(b) $I_m = \frac{V_m}{R} = \frac{25\,V}{10\,\Omega} = 2.5\,A$

v & i are in phase,

$i = 2.5 \sin(377t + 60^\circ)$



Example 02: The current through a 0.1 H coil is provided. Find the sinusoidal expression for the voltage across the coil. sketch v & i.

(a) $i = 10 \sin 377t$

(b) $i = 7 \sin (377t - 70°)$

Sol:-

(a) $X_L = \omega L = (377 \text{ rad/s})(0.1 \text{ H})$

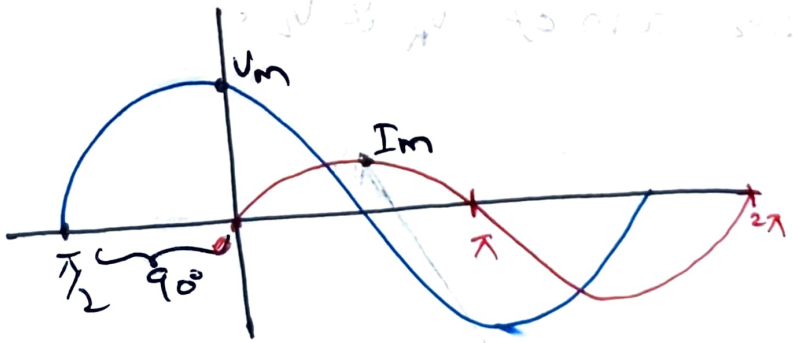
$= 37.7 \Omega$

$V_m = I_m X_L = (10 \text{ A})(37.7 \Omega)$

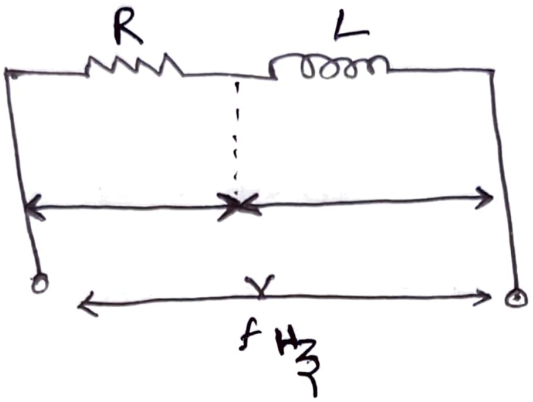
$= 377 \text{ V}$

for a coil v leads i by 90°. Therefore

$v = 377 \sin (377t + 90°)$



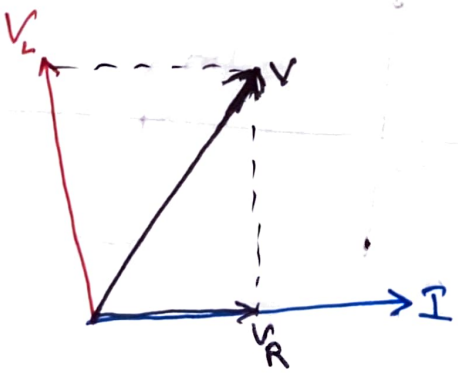
Inductance & Resistance in series:

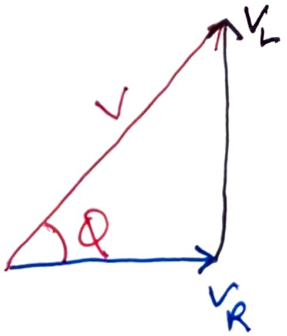


→ The p.d across 'R' is in phase with current through it ($\phi = 0°$)

→ The p.d across the inductor will lead current by 90°. ($\phi = 90°$).

→ The total voltage applied, V, will be the phasor sum of V_R & V_L.





$$V^2 = V_R^2 + V_L^2$$

$$\Rightarrow (IZ)^2 = (IR)^2 + (IX_L)^2$$

dividing the above equation with I^2

$$Z^2 = R^2 + X_L^2$$

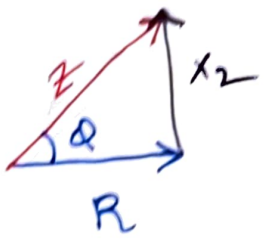
$$\Rightarrow Z = \sqrt{R^2 + X_L^2}\ \Omega$$

From Impedance triangle:

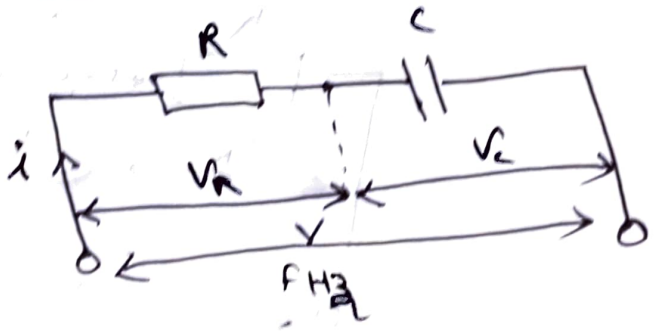
$$\cos\phi = \frac{R}{Z} = \frac{V_R}{V}$$

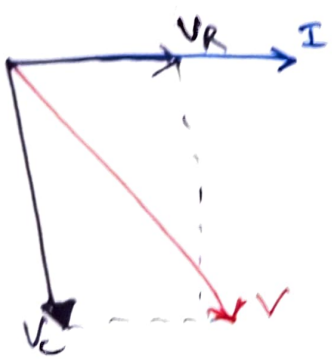
$$\sin\phi = \frac{X_2}{Z} = \frac{V_L}{V}$$

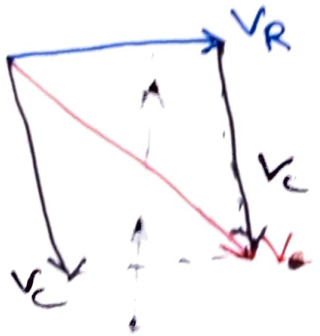
$$\tan\phi = \frac{X_2}{R} = \frac{V_L}{R}$$



Resistance and capacitance in series:



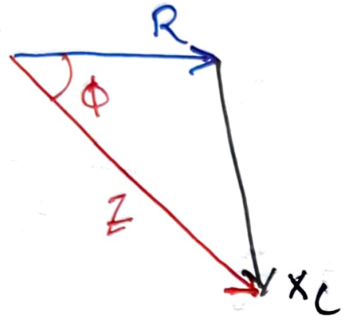




$$(IZ)^2 = (IR)^2 + (IX_C)^2$$

$$\Rightarrow Z^2 = R^2 + X_C^2$$

$$\Rightarrow Z = \sqrt{R^2 + X_C^2}\ \Omega$$

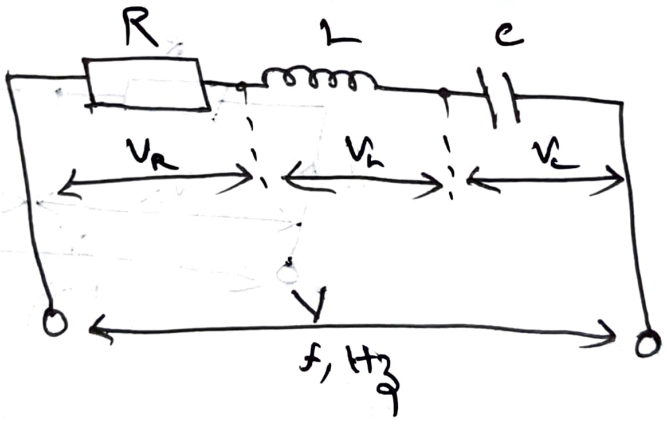


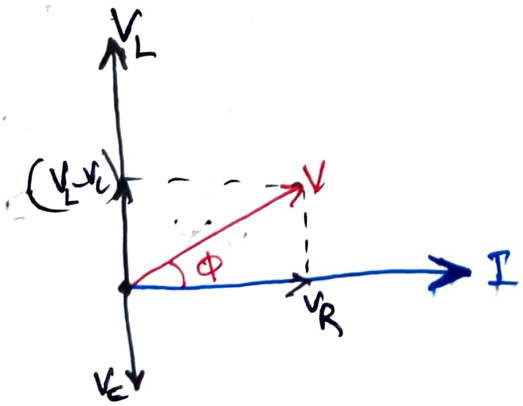
$$\phi = \cos^{-1}\left(\frac{R}{Z}\right)$$

$$\sin\phi = \frac{X_C}{Z} = \frac{V_C}{V}$$

$$\tan\phi = \frac{X_C}{R} = \frac{V_C}{V_R}$$

Resistance, Inductance and capacitance in series.



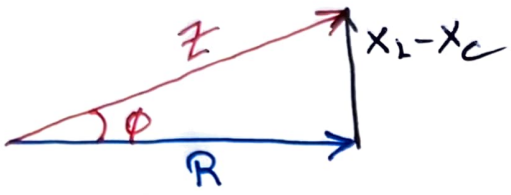


$$V^2 = V_R^2 + (V_L - V_C)^2$$

$$\Rightarrow (IZ)^2 = (IR)^2 + (IX_L - IX_C)^2$$

$$\Rightarrow Z^2 = R^2 + (X_L - X_C)^2$$

$$\Rightarrow \quad Z = \sqrt{R^2 + (X_2 - X_C)^2} \ \Omega$$



$$\cos\phi = \frac{R}{Z}; \qquad \sin\phi = \frac{X_2 - X_C}{Z}$$

$$\tan\phi = \frac{X_2 - X_C}{R}$$